

The Impact of Competitive Intelligence Services on Online Marketplaces

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Recent innovations have driven a steady increase in online marketplace transactions. To remain competitive, numerous marketplace platforms and independent data providers offer *Competitive Intelligence Services* (CIS), enabling sellers to explore not only their market potential but also that of their competitors. In this paper, we employ a game-theoretical approach to competitive learning to analyze the impact of CIS on participants with varying market shares in an online marketplace. In the presence of noisy demand signals, the online platform benefits from offering CIS to both sellers. This is because high demand noise makes demand exploration difficult for each seller in the first period. Consequently, price competition under poor knowledge of the price-demand relationship in the second period leads to a lower payoff for each seller as well as the platform. However, as demand uncertainty decreases, the platform prefers inducing CIS exclusively for the seller with the larger market share. This scenario leads to *signal-jamming* behavior between the sellers, which results in a win-win-win situation for both sellers and the platform provider. Finally, we consider various model extensions and discuss the managerial implications for the design and regulation of competitive intelligence services in online marketplaces. test

Key words: Online marketplaces; Platform economy; Competitive learning; Mechanism design; Signal-jamming; Price competition

1. Introduction

Online marketplaces have witnessed significant growth over the years. In 2020, these platforms accounted for more than 18% of global retail sales, and this figure is expected to reach 21.8% in 2024 (Coppola 2021). Online marketplaces differ considerably within the retail sector, depending on whether they facilitate business-to-business (B2B), business-to-consumer (B2C), or peer-to-peer (P2P) transactions (Anastasia 2020); whether they exchange goods, services, or information (Pryhodko 2020); and whether they generate revenue via advertisements, sales commission, or subscription services. Regardless, the value proposition of all online platforms is the same: facilitating transactions between independent supply and demand-side participants (Täuscher and Laudien 2018). As such, their growth will continue in the foreseeable future, and they will

continue to attract not only more consumers but also more *sellers*, thereby creating a win-win scenario for all stakeholders.

In order to gain a competitive advantage, many sellers on online marketplaces employ tools that gather, manage, and use information collected from not only their own customers but also from their competitors. In fact, a survey conducted by Forrester Consulting (Forrester 2016) among 115 technology, marketing, and business professionals suggests that exploring the customer journey in a competitive customer-facing web environment is the highest strategic priority for sellers. However, not all online sellers have the resources to carry out competitive intelligence activities on their own. According to a 2018 survey of mostly U.S.-based Amazon sellers, 73% of seller firms had only one to five employees (and 93% had fewer than 50 employees)¹. This is why easy-to-use Competitive Intelligence² Services (CIS), powered by artificial intelligence and machine learning, have gained popularity on platforms like Google, Amazon, and others (Mroz 2018). For example, subscribing to Amazon Marketing Web Services (recently revised to Selling Partner API) enables sellers to understand the source of traffic for not only their own products but also for those of their competitors (Malachard 2020). WeChat Marketing Services allow sellers to segment their own customers as well as those of their competitors based on clickstream data (TDA 2019). The need for competitive intelligence services has given rise to a number of third-party data brokers, such as Scrapfly, Competitive Analytics, Liberty Metrics, and Jungle Scout, who provide automated data collection services to sellers in online marketplaces.

However, even though CIS are widely popular, little is known about how these services, provided by the platform to its sellers, affect seller decisions and platform actions over time. On the one hand, consider the decisions taken by the sellers after receiving business intelligence inputs. If, for instance, all the sellers on a platform are subscribed to CIS, then they will all have access to intelligence inputs. Undoubtedly, these inputs will alter their decisions compared to a situation in which only a few sellers are subscribed to CIS. In the latter case, only the sellers receiving CIS inputs will gain a competitive advantage, whereas in the former case, the effect of CIS is likely to be eroded. On the other hand, consider the decisions taken by sellers during the data collection/analysis stage, i.e., before the availability of business intelligence. In that stage, the decisions of all sellers, whether or not they have subscribed to CIS, are likely to directly affect the quality of information gathered from the data collected. Thus, sellers have an opportunity to increase or decrease the quality of intelligence available to the platform. Overall, after considering the ex-ante and ex-post impacts of CIS on all the stakeholders, we arrive at the following research questions:

- How does offering CIS to both sellers affect the payoffs of the platform participants and consumer welfare?

¹ <https://www.statista.com/statistics/886904/amazon-seller-business-size-by-employees/>

² Competitive intelligence means understanding and learning about what is happening outside the firm to enhance its competitiveness. It entails gathering as much information as possible, as quickly as possible, about the external environment, which includes the industry as a whole and relevant competitors.

- What is the impact of offering CIS exclusively to only one seller on the decisions and payoffs of the platform, sellers, and consumers?

To address these research questions, we develop a stylized two-period model that studies two asymmetric sellers (in terms of their market potentials) who sell their products through an online marketplace, which we refer to as the platform³. The platform charges a fixed fee for CIS, and the sellers decide whether to subscribe to the CIS or not. The sellers then compete in prices in both periods. In order to capture the value and impact of CIS, we employ a demand-learning framework. Specifically, the market potential (represented by online traffic) of each seller is originally unknown to both sellers and the platform provider. In the first period, each seller sets an exploratory price with the objective of learning the true market potential during the first period. At the end of the first period, each seller observes only their own sales, whereas the platform observes sales of both sellers, and therefore has access to superior information. Based on the observed sales, both sellers and platform update their beliefs about the true market potential. If a seller is subscribed to CIS, then, at the end of the first period, that seller receives the updated belief of the platform. Otherwise, she relies only on her own updated belief and decides on the second period price. Building on the above modeling framework and considering the possibility of each seller's subscription decision for CIS information, we create four scenarios and provide a complete characterization of the equilibrium prices and payoffs in both periods.

To address the first research question, we consider two scenarios: one where both sellers subscribe to the CIS, and the other where neither seller does. Under each scenario, we characterize the subscription fees that are incentive compatible for the sellers. Finally, we compare the equilibrium payoffs from the platform's perspective. Our analysis shows that if the observations from the first period sales are too uncertain for the sellers to accurately identify the true market potential or when the degree of competition between the sellers is low, the platform has an incentive to induce both sellers to subscribe to the CIS. When either of these conditions is not met, the platform actually prefers not to induce any of the sellers to subscribe to the CIS. Consequently, each seller has to find her own balance between exploring the demand parameters and exploiting the available information in a competitive environment. This result is robust to various extensions under which seller asymmetry is observed in marginal cost, the total parameters governing the market potential become state dependent, and the demand noise for each seller is non-uniform and partially correlated.

In order to address the second research question, we consider scenarios in which one seller subscribes to CIS while the other seller does not. Our analysis shows that the platform finds it incentive compatible to induce only the seller with larger market to subscribe, accomplishing this by setting a subscription fee that makes it profitable for only the larger-market seller to participate. Analyzing the equilibrium, contrary to our intuition, reveals that a seller with a smaller market share may indeed benefit from not subscribing to CIS. This counterintuitive finding arises due to the nature of first-period price competition between the

³ Although technically and legally the marketplace provider and the platform owner may be different entities we do not make this distinction and assume that platform owner is also the marketplace provider who operates the platform for sellers and buyers.

sellers when only one of them subscribes to CIS. Specifically, the seller who receives updated information about the market potential from the platform has an incentive to charge a price that makes it difficult for the competitor to explore. We call this *signal-jamming* and analyze its impact on equilibrium prices. Interestingly, signal-jamming leads to both sellers charging higher first-period prices, which turns out to be mutually beneficial for all parties involved, resulting in a win-win-win scenario for both sellers and the platform provider. However, from a consumer welfare perspective, this scenario hurts the customers because of the high prices at equilibrium.

Our findings have managerial implications for platforms and regulators. First, our study identifies when and with whom marketplace providers should offer CIS: when the demand information is too uncertain for the sellers, the platform should step in and induce both sellers to subscribe to the CIS by offering this service. Our results show that this would prevent both sellers from charging the suboptimal prices and increase the profit for the whole platform. We expect that products with short-life cycles, such as fashion items or high-tech products, are likely to have more uncertain demand; hence, in such cases, the platform should take the initiative to explore and share the demand *symmetrically* with all marketplace players. However, in the case of products with relatively less uncertain demand, such as commodity items or non-seasonal products, our results show that the platform should either delegate the demand exploration to each seller or, if it must interfere, induce only the seller with the larger market share to subscribe to the CIS. That being said, from a regulator's perspective, this asymmetric information sharing leads to lower consumer welfare. As such, a marketplace regulator should ensure that platforms do not offer a subscription policy that affects marketplace players asymmetrically. In fact, the recent data-sharing regulations designed by the European Union aim to create a level playing field for all the stakeholders in online marketplaces⁴. Warnings have been issued to ensure proper regulation of platforms and guard against their powerful network effects (Sokol and Van Alstyne 2021).

The rest of this paper is organized as follows. In § 2, we position our work in the context of other relevant literature. In § 3, we develop our modeling framework. In § 4, we analyze the impact of offering CIS to both sellers. In § 5, we analyze the impact of offering CIS exclusively to only one of the sellers. We extend our results in § 6 and conclude the paper with managerial and theoretical insights in § 7.

2. Related Literature

The design of initiatives for information exchange among members of the distribution channel has always been a crucial topic (Guo and Zhao 2009, Gal-Or et al. 2008, He et al. 2008, Gal-Or et al. 2007, Dukes et al. 2017). In the modern era, this has become even more critical due to the ability of online platforms to efficiently collect extensive data on consumers' purchasing habits. Consequently, the strategies for deciding to whom and how this information is shared, often hidden from upstream manufacturers and rival sellers, have

⁴ See, for example, the Digital Markets Act(Commission 2020) of the European Commission dated Dec. 15, 2020.

gained greater significance (Zha et al. 2023, Shi et al. 2023, Long et al. 2022, Caldieraro et al. 2018). Our paper contributes to this research stream by examining how platforms can benefit from competitive learning through the governance of Competitive Intelligence Services (CIS) among participants with varying market shares in an online marketplace.

The platform's strategy for sharing sales information is based on how this information influences price competition among retailers and, consequently, the profit of each party, as well as the overall channel profit (Lai et al. 2022). In reality, sellers may not have precise knowledge of the relationship between price and demand. However, they can learn it through price experimentation (exploration) without completely sacrificing their revenue maximization goal (exploitation). This exploration-exploitation trade-off has been examined in prior research (Chen and Chen 2015). With this in mind, our primary goal is to provide context for our work and highlight its value and relevance. Accordingly, we categorize existing research into three groups based on the market structures they analyze: (i) a monopolistic seller, (ii) competing sellers, and (iii) a platform and competing sellers.

Early work on the exploration-exploitation trade-off analyzed the conditions under which a monopolistic seller would eventually obtain complete information about the underlying demand environment, while it focused only on exploitation (Rothschild 1974, McLennan 1984, Easley and Kiefer 1988, Aghion et al. 1993, Harrington 1995). Later, the focus shifted to developing algorithms with provable worst-case bounds based on minimizing regret. One approach to categorizing this stream of literature is by the mechanism used to estimate the demand parameters: Bayesian updating (Narayan et al. 2011, Lin 2006, Araman and Caldentey 2009, Gallego and Talebian 2012, Ching et al. 2013), realization probability (Cao and Zhang 2021), and linear least squares (Misra et al. 2019, Bertsimas and Perakis 2006). More recent work considers a variety of revenue management issues, such as a limited range of prices available to the seller (Perakis and Singhvi 2019), customized pricing decisions (Chen and Iyer 2002, Liu and Zhang 2006, Chen and Gallego 2018), and the impact of markdown policies for demand learning with forward-looking customers (Birge et al. 2019). Our work differs from the monopoly stream in that we study seller competition. We identify the subgame-perfect equilibrium prices, which take into account the optimum exploration-exploitation trade-off given the competitive dynamics between sellers and the platform.

Accounting for competition in a demand-learning framework not only requires a rich underlying demand model and a convoluted game-theoretic analysis but also complicates the exploration-exploitation trade-off faced by the sellers. Each seller needs to decide on the level of exploration while being cognizant that their experimentation can benefit rival sellers by increasing their profits or by providing them with free demand information (Bolton and Harris 1999). The majority of the works in this stream deploy a two-period model with two sellers. Sellers have imperfect information about the demand model when making their first-period decisions. In the second period, sellers make use of Bayesian updating based on the observed decisions and outcomes. This research focuses on identifying how the subgame-perfect equilibrium decisions compare

with the myopic case (no exploration), the monopoly case, or the case of perfect demand information. Thus, the most relevant categorization of the existing work from our perspective is based on the distribution of information among the stakeholders. Specifically, it is either assumed that all sellers observe all the sales and decisions (Mirman et al. 1994, Alepuz and Urbano 1999, Dimitrova and Schlee 2003, Harrington 1995, Belleflamme and Bloch 2001, Keller and Rady 2003, Tsunoda and Zennyo 2021) or that the first-period decisions (output in Cournot models and price in Bertrand models) are private information (Riordan 1985, Mirman et al. 1993, Bernhardt and Taub 2015, Bonatti et al. 2017, Stern and Birge 2020). When first-period decisions are common knowledge, sellers tend to adjust their decisions to acquire more information at the end of the first period. However, when first-period decisions are private information, an opportunity arises to manipulate the information available to opponents, i.e., signal-jamming. The salience of this work lies in its use of a platform (i.e., online marketplace) through which all the sellers compete. As an entity that has access to the sales individually observed by each seller, the platform can also explore the true demand by itself and share this information with all or none of the sellers or even exclusively with one seller, which is a unique and novel scenario. By explicitly considering platform-mediated interactions, we explore the interaction between the platform's exploration and sellers' pricing decisions.

Our model is also related to Operations Management (OM) literature that explores the impact of sharing demand information in supply chains (Chen 2003, Wang et al. 2022, Guo et al. 2014, Jain 2022). Previous studies often examine bilevel supply chain structures and involve vertical (Cachon and Lariviere 2001) or horizontal competition (Li and Zhang 2008, Ha and Tong 2008), as well as information-sharing games among competing supply chains (Ha et al. 2011, Shamir and Shin 2016). The credibility of shared demand information is a central concern in this literature, addressed through screening (Ha and Tong 2008) or signaling mechanisms (Gümüő 2014). Our work differs in two main ways. First, while many studies assume one supply chain partner possesses *ex-ante* private information about demand, in our model, demand information is *ex-ante* unknown to all parties. Second, prior research often assumes that private information remains unaffected by other supply chain firms' decisions. In contrast, our model considers how each party's (sellers and platform) knowledge of true demand information is influenced by competing firms' (pricing) decisions. These unique features allow us to analyze the exploration-exploitation trade-off in the context of platform-mediated competition.

Recent works point to a growing interest in modeling the exploration-exploitation trade-off over a duration longer than two periods (Bertsimas and Perakis 2006, Perakis and Sood 2006, Stern and Birge 2020). Perakis and Sood (2006) propose a robust optimization approach that maximizes the revenue for each seller under the most adverse instances of unknown parameters of the demand function. They prove the existence of equilibrium policies and develop an iterative learning algorithm for finding them. In the work by Stern and Birge (2020), each seller estimates the underlying demand curve using least squares estimation. Interestingly, the results suggest that sellers may prefer to stay ignorant, as being informed may reduce prices (ignorance

may lead to collusion). The analysis of a strategic game shows the existence of an equilibrium in which firms actively avoid learning (by avoiding experimentation) the true value of demand and attain a collusive outcome even in finite-time horizons.

Our work differs from the literature on information sharing in retail platforms. First, unlike much of the literature (e.g., Bimpikis et al. (2019); Liu et al. (2021); Wang et al. (2022); Zha et al. (2023)), we do not assume that the platform or sellers initially hold private demand information; instead, we introduce a novel self-learning channel whereby sellers can infer demand from their own sales. Second, while most existing studies adopt a single-period or single-decision setting in which uninformed parties remain passive, we allow sellers to make two sequential pricing decisions, which creates a new intertemporal trade-off between exploration and exploitation. Finally, relative to empirical work such as Huang et al. (2022), which studies adaptive learning in real markets, our contribution lies in a game-theoretic analysis of how subscription-based and self-learning channels jointly shape equilibrium pricing, information acquisition, and welfare outcomes.

Last but not least, our model is also related to the literature on Bayesian persuasion and strategic information design Kamenica and Gentzkow (2011). In our model, the platform effectively commits to an information disclosure policy-implemented through its CIS subscription design-that shapes sellers' posterior beliefs and pricing behavior. Unlike classical persuasion models, however, information in our setting is endogenously generated through sellers' pricing decisions and sales realizations, and competing sellers can strategically manipulate the informativeness of this data. This interaction gives rise to signal-jamming incentives that are unique to platform-mediated competition and highlight new limits and possibilities of information design in competitive marketplaces.

3. Model Framework

Consider two sellers, A and B , who are selling their products through an online platform, O , over two periods. Both sellers are uncertain about their market size. To capture this uncertainty, we assume that the market size of each seller depends on the demand state θ , where θ can be high (i.e., $\theta = H$) or low (i.e., $\theta = L$) with equal probability. The true value of the demand state is set by nature and is a priori unknown to both sellers and the platform. We assume that sellers A and B vary in terms of their potential market sizes. Let $\alpha_i(\theta)$ denote the market size of seller i , where $i \in \{A, B\}$ depends on the demand state θ . Without loss of generality, we assume that seller A 's market size is stochastically larger than that of seller B , i.e., $\alpha_A(\theta) \geq \alpha_B(\theta)$ for any realization of the demand state $\theta \in \{H, L\}$.

Note that, at the end of the first period, each seller observes its own sales, while the platform observes the sales of both sellers. This highlights the significance of CIS benefits; the platform has a comprehensive view of total sales, resulting in a more accurate updated belief about the demand state. Consequently, the platform can share its updated belief with the seller(s) who have subscribed to the CIS. To model this, we define ϕ to represent the subscription fee; a lump-sum amount required to be paid by sellers if they desire access to CIS benefits. Let $y_i \in \{0, 1\}$ represent seller- i 's decision to subscribe to CIS (i.e., $y_i = 1$) or not (i.e., $y_i = 0$).

Consequently, four subscription regimes emerge: (i) regime $J = \emptyset$ under which neither of the sellers subscribe (i.e., $y_A = y_B = 0$), (ii) regime $J = AB$ under which both sellers subscribe (i.e., $y_A = y_B = 1$), (iii) regime $J = A$ under which only the big seller subscribes (i.e., $y_A = 1; y_B = 0$), and (iv) regime $J = B$ under which only the small seller subscribes (i.e., $y_A = 0; y_B = 1$). In order to avoid trivial cases, in which a seller can always learn the true demand state, we introduce a random shock in the demand function in a multiplicative form⁵ via a random variable ϵ_t for $t \in \{1, 2\}$, where ϵ_t is independent and identically distributed uniformly $\epsilon_t \sim U[0, 1]$. The sales observed by each seller in period t under regime J have the following linear functional form⁶:

$$d_{i,t} \left(p_{i,t}^J, p_{-i,t}^J \right) = \alpha_i(\theta) - p_{i,t}^J + 2\gamma\epsilon_t(p_{-i,t}^J - p_{i,t}^J)$$

where $p_{i,t}^J$ and $p_{-i,t}^J$ denote the prices set by seller i and the competitor $-i$, respectively, in period t , given the subscription regime $J \in \{AB, A, B, \emptyset\}$. Note that parameter γ controls the magnitude of both own- and cross-price elasticities, and as such, influences the degree of price competition between the sellers. When γ is set to zero, the cross-elasticity term vanishes, and therefore, the price set by the competitor firm has no effect on the demand for the firm's product. However, as γ increases, a change in the competitor's price has a progressively greater impact on the firm's demand curve. Note also that the absolute value of the own-price elasticity of the demand function is greater than that of the cross-price elasticity in the expected sense, which is consistent with the findings in economics literature (e.g. Vives 1999).

We assume that the marginal cost of the product is normalized to zero and consider the non-zero marginal cost in §6.1. Given the subscription decisions $y_i, y_{-i} \in \{0, 1\}$, which result in regime J , the expected channel profit generated by seller i can be expressed as follows:

$$\pi_{Cli}^J \left(P_i^J, P_{-i}^J \right) = E_{\theta, \epsilon_t} \left[\sum_{t=1}^2 d_{i,t}^J \left(p_{i,t}^J, p_{-i,t}^J \right) p_{i,t}^J \right]$$

where P_i^J represents the price trajectory of seller i over two periods; i.e., $P_i^J \equiv (p_{i,1}^J, p_{i,2}^J)$, the expectation is taken with respect to the a priori distribution on the demand state θ and the multiplicative noise ϵ_t for $t \in \{1, 2\}$. Consequently, the payoff of seller i can be written as follows:

$$\pi_i^J \left(P_i^J, P_{-i}^J \right) = (1 - \beta)\pi_{Cli}^J \left(P_i^J, P_{-i}^J \right) - y_i\phi \quad (1)$$

where β denotes the commission rate charged by the platform⁷ for each unit sold. To induce the information regime $J \in \{AB, A, B, \emptyset\}$, the platform needs to solve the following optimization problem:

$$\max_{\phi} \pi_O^J = \beta \sum_{i \in \{A, B\}} \pi_{Cli}^J \left(P_i^J, P_{-i}^J \right) + \sum_{i \in \{A, B\}} y_i\phi \quad (2)$$

$$\text{Subject to } \pi_i^J \left(P_i^J, P_{-i}^J \right) \geq \pi_i^{\check{J}} \left(P_i^{\check{J}}, P_{-i}^{\check{J}} \right), \check{J} \neq J, i \in \{A, B\} \quad (3)$$

⁵ Both multiplicative and additive cases are extensively considered in the literature. We refer to Petruzzi and Dada (1999) for a detailed discussion on this issue.

⁶ Consistent with game theory terminology, we denote the competing seller by subscript $-i$ throughout the paper.

⁷ For instance, according to Amazon pricing policy (<https://sell.amazon.com/pricing>), Amazon charges a fixed commission fee of 15% for categories like backpacks, handbags and luggage, as well as toys and games, 17% for clothing and accessories, and 8% for consumer electronics such as cell phone devices and computers.

where constraint (3) represents the incentive compatibility constraint, ensuring that seller i is better off by choosing a subscription decision aligned with regime J . The detailed account of events is depicted in Figure 1. At the outset, nature determines the true demand state $\theta \in \{H, L\}$. The game is then played in the following stages:

- *Subscription stage*: Without knowledge of the value of θ , the platform sets the CIS subscription fee ϕ . Sellers then rely on their prior beliefs to decide whether to subscribe to CIS (i.e., $y_i = 1$) or not (i.e., $y_i = 0$). Consequently, the subscription regime $J \in \{AB, A, B, \emptyset\}$ is established and remains valid for two periods. A key practical consideration is whether, in the presence of CIS services, a seller consents to the platform sharing her sales information with the other seller⁸. In §4, we characterize the equilibrium under subscription regimes AB and $\emptyset$. Under regime AB , the platform offers the CIS subscription to both sellers, and it is implemented only if both sellers provide their consent. In §5, we extend the analysis to subscription regimes A and B , where the CIS subscription is offered unilaterally to a single seller, and thus consent from the other seller is required.
- *Price competition in period 1*: Sellers A and B use their a priori beliefs about the true demand state and simultaneously set their prices $p'_{A,1}$ and $p'_{B,1}$, respectively. At the end of period 1, each seller observes its own sales, $d_{i,1}, i \in \{A, B\}$, while the platform observes the total sales; $d'_{A,1}$ and $d'_{B,1}$.
- *Belief revision stage*: The platform updates its belief θ and learns the true demand state; i.e., $Pr(\theta_O = \theta) = 1$. It shares these updated beliefs θ_O with seller(s) if they opted for subscription to the CIS. Specifically, let θ_i denote the posterior belief of seller $i \in \{A, B\}$ regarding demand at the end of the first period. Therefore, $\theta_i = \theta_O$ if $y_i = 1$.
- *Price competition in period 2*: Seller A and B use their posterior beliefs θ_i regarding the true demand state and simultaneously set their prices $p'_{A,2}$ and $p'_{B,2}$, respectively. The sales $d'_{A,2}$ and $d'_{B,2}$ would be observed at the end of the second period.

To simplify the notation, in our baseline analysis, we assume that the total market size is normalized and equal to 1 under both demand states⁹. Accordingly, we use $\alpha_A(\theta) = \alpha_\theta$ and $\alpha_B(\theta) = 1 - \alpha_\theta$ for $\theta \in \{H, L\}$.

3.1. First-Best Solution

In this section, we analyze the *first-best solution* assuming that the platform controls the pricing decision of both products¹⁰. Let $p_{i,t}^{fb}$ denote the first-best prices charged by the platform for product of seller $i \in \{A, B\}$ and in period $t \in \{1, 2\}$. Then, the platform solves the following optimization problem:

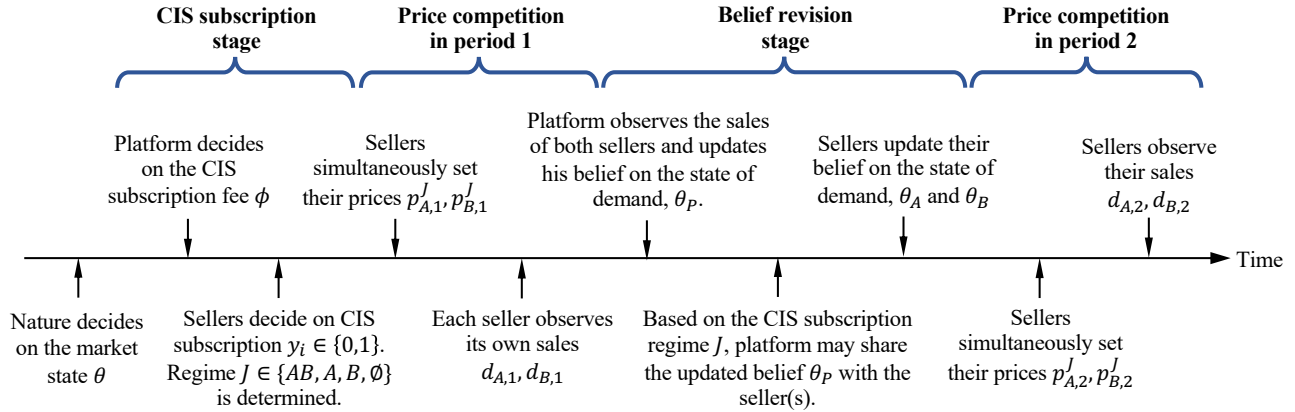
$$\max_{p_{A,t}^{fb}, p_{B,t}^{fb}} \pi_O^{fb} = \sum_{t \in \{1, 2\}} \pi_{O,t}^{fb}(p_{A,t}^{fb}, p_{B,t}^{fb}) \quad (4)$$

⁸ We assume that if a seller subscribes, the seller will provide consent. However, if a seller chooses not to subscribe to the CIS, she may either provide consent or withhold it from the platform.

⁹ We relax this assumption in §6.

¹⁰ In the first best solution, seller A and B sell differentiated products and refer to them as product A and B .

Figure 1 Sequence of events



The optimal solution is fully characterized in the following proposition. (Note that the proofs for all propositions are provided in the Appendix [Section 1](#).)

PROPOSITION 1. *Under the first-best scenario:*

- In the first period, the platform sets $p_{A,1}^{fb} = \bar{p}^{fb} + z_p^{fb} \Delta$ and $p_{B,1}^{fb} = \bar{p}^{fb} - z_p^{fb} \Delta$ where $\Delta \equiv \frac{\alpha_H - \alpha_L}{4\gamma}$, $\bar{p}^{fb} = \frac{1}{4}$, and $z_p^{fb} = \frac{\gamma}{1+2\gamma} \frac{\alpha_H + \alpha_L - 1}{\alpha_H - \alpha_L}$,
- The learning is perfect; the platform learns the true demand state, i.e., $\theta_O = \theta$,
- In the second period, the platform charges the state-dependent prices, i.e.,

$$p_{A,2}^{fb} = \frac{1}{4} + \frac{2\alpha_\theta - 1}{4(1+2\gamma)}, \quad \text{and} \quad p_{B,2}^{fb} = \frac{1}{4} - \frac{2\alpha_\theta - 1}{4(1+2\gamma)}$$

Recall that, by assumption, seller A's product is expected to achieve a higher market share than seller B's product. Therefore, both first- and second-period prices for product A are set higher than those for product B. This price distinction is evident in the optimal price structure described in Proposition 1, where the price for product A surpasses that of product B. Furthermore, the price difference between the products diminishes as the degree of substitution (measured by γ) increases. This is intuitive because the platform prefers to narrow the price gap to prevent demand cannibalization between the products.

3.2. Preliminaries

Before characterizing equilibrium under different CIS regimes J , we first formulate the problem and define the solution concept for the equilibrium. Under a specific CIS regime, each seller sets her price in each period by solving the following price optimization problem:

$$\max_{p_{i,t}} \pi_i^J = \pi_{i,1}^J(p_{i,1}, p_{-i,1}) + \pi_{i,2}^J(p_{i,2}, p_{-i,2}) \quad (5)$$

given that $p_{-i,t}$ solves the competing seller's optimization problem:

$$\max_{p_{-i,t}} \pi_{-i}^J = \pi_{-i,1}^J(p_{-i,1}, p_{i,1}) + \pi_{-i,2}^J(p_{-i,2}, p_{i,2}) \quad (6)$$

We utilize the concept of Perfect Bayesian Equilibrium (PBE) (Fudenberg and Tirole 1991) to solve the game described above. Given the subscription regime J , the PBE necessitates the satisfaction of the following two conditions:

- Subgame perfection condition: Given the updated (i.e., a posterior) beliefs θ_A and θ_B , there is no profitable deviation for any seller from the equilibrium prices sequentially in the sense that:

- In period $t = 2$, there is no profitable deviation for a seller given the equilibrium first-period prices and the competitor's second-period price.

- In period $t = 1$, there is no profitable deviation for a seller given the equilibrium subscription decisions, and that sellers follow their best responses in period $t = 2$.

- Bayesian updating condition: Given the first-period prices, sellers update their beliefs about the true demand state in line with the Bayesian updating rule.

4. Model Analysis

In this section, our focus is on the competitive setting, and we analyze the impact of inducing both sellers to subscribe or not to subscribe to CIS on their equilibrium decisions and payoffs. To accomplish this, we first examine the equilibrium in the case of $J = AB$ in §4.1, where both sellers choose to subscribe to CIS. Subsequently, in §4.2, we characterize the equilibrium prices in the case of $J = \emptyset$, where neither of the sellers opts to subscribe to CIS. Next, in §4.3, we characterize the full equilibrium and evaluate the impact of the equilibrium on payoffs.

4.1. Regime $J = AB$: Inducing Both Sellers to Subscribe to CIS

In this case, the platform can induce both sellers to subscribe to CIS by finding a subscription fee ϕ in Eq. (2) that satisfies the following two incentive compatibility constraints:

$$\pi_B^{AB}(P_B^{AB}, P_A^{AB}) \geq \pi_B^A(P_B^A, P_A^A) \quad (7)$$

$$\pi_A^{AB}(P_A^{AB}, P_B^{AB}) \geq \pi_A^B(P_A^B, P_B^B) \quad (8)$$

The above constraints collectively ensure that both sellers are better off by subscribing to CIS, thus forming the information regime $J = AB$. Moreover, since the platform has access to both sellers' sales data and can always identify the true demand state at the end of the first period, both sellers are guaranteed to learn the true state of demand in the first period. This means that, in contrast to Proposition 4, where equilibrium first-period prices were characterized under an exploration-exploitation trade-off, in the presence of CIS, the sellers focus purely on *exploitation*, i.e., maximizing their first-period profit through price competition.

PROPOSITION 2. *The platform can induce subscription regime $J = AB$ only if $\alpha_H \in (\alpha_L, \alpha_H^{AB}]$, where α_H^{AB} is the solution of $\pi_{CIB}^{AB}(P_B^{AB}, P_A^{AB}) - \pi_{CIB}^A(P_B^A, P_A^A) = 0$. The CIS subscription is offered at a cost $\phi^{AB} = (1 - \beta)[\pi_{CIB}^{AB}(P_B^{AB}, P_A^{AB}) - \pi_{CIB}^A(P_B^A, P_A^A)]$; both sellers subscribe and set the following equilibrium prices for the first period:*

$$p_{A,1}^{AB} = \bar{p}^{AB} + z_p^{AB} \Delta, \quad \text{and} \quad p_{B,1}^{AB} = \bar{p}^{AB} - z_p^{AB} \Delta,$$

where $\bar{p}^{AB} = \frac{1}{2(2+\gamma)}$, $z_p^{AB} = \frac{2\gamma(\alpha_H + \alpha_L - 1)}{(2+3\gamma)(\alpha_H - \alpha_L)}$, and $\Delta = \frac{\alpha_H - \alpha_L}{4\gamma}$. After the first period, the platform learns the true demand state, i.e., $\theta_O = \theta$, and shares the information with both sellers, i.e., $\theta_A = \theta_B = \theta$. In the second period, sellers set the following equilibrium prices:

$$p_{A,2}^{AB} = \frac{\gamma + (2+\gamma)\alpha_A(\theta)}{(2+\gamma)(2+3\gamma)}, \quad \text{and} \quad p_{B,2}^{AB} = \frac{\gamma + (2+\gamma)\alpha_B(\theta)}{(2+\gamma)(2+3\gamma)}.$$

The main takeaways from Proposition 2¹¹ are as follows. First, to induce both sellers to subscribe, the platform sets the subscription fee high enough to capture the entire surplus that the smaller seller would gain from subscribing to CIS. However, the larger seller, i.e., seller A can still retain a portion of the surplus after accounting for the subscription fee. Second, inducing both sellers to subscribe to CIS is only feasible when the difference between the demand states under the a priori distribution is sufficiently low (i.e., $\alpha_L < \alpha_H \leq \alpha_H^{AB}$). Note that the first-period price of seller A (and seller B) increases (decreases) with α_H . When this difference becomes significant, i.e., $\alpha_L < \alpha_H^{AB} < \alpha_H$, it becomes infeasible for the platform to satisfy the incentive compatibility constraint (7). This is because the price competition intensifies in the first period; as α_H increases, seller A becomes more optimistic about a favorable market state, leading her to raise her price to take advantage of the opportunity. To mitigate the price competition, seller B prefers not to subscribe, opting to wait, observes her own sales, and makes decisions based on her posterior beliefs in the second period. Lastly, under regime $J = AB$, the average equilibrium price in the first period is lower than the first-best average price in Proposition 1 (i.e., $\bar{p}^{AB} < \bar{p}^{fb}$). However, the difference between equilibrium prices is higher than that of the first-best prices (i.e., $z_p^{AB} > z_p^{fb}$). These results are due to the price competition, leading to lower and more differentiated price levels compared to the first-best scenario, where the platform controls sellers' prices.

4.2. Regime $J = \emptyset$: Inducing No subscription to Both Sellers

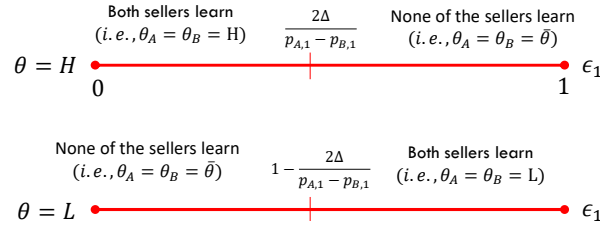
In this section, we analyze the scenario under which the platform induces both sellers not to subscribe to the CIS. To achieve this, the platform needs to satisfy the following incentive compatibility constraints:

$$\pi_B^\emptyset(p_B^\emptyset, p_A^\emptyset) \geq \pi_B^B(p_B^B, p_A^B) \quad (9)$$

$$\pi_A^\emptyset(p_A^\emptyset, p_B^\emptyset) \geq \pi_A^A(p_A^A, p_B^A) \quad (10)$$

The above constraints collectively ensure that both sellers are better off by not subscribing to CIS, thus forming the information regime $J = \emptyset$. Note that, without CIS, sellers only observe their own sales at the end of the first period. Using backward induction, we first establish the beliefs regarding the true demand state for each seller based on their first-period prices. Depending on the ordering of first-period prices between sellers A and B, two cases arise: (i) $p_{A,1} \geq p_{B,1}$ and (ii) $p_{A,1} \leq p_{B,1}$. Throughout the paper, we focus on the first case, because, as shown in the Appendix, it is the only scenario that can be sustained on the equilibrium path.

¹¹ Proof in Appendix Section 2

Figure 2 Sellers' belief updating based on first-period prices and demand shock

Given the price difference in the first period and realized first-period demand, one can characterize the belief revision of each seller.

PROPOSITION 3. *Without subscribing to CIS, seller i updates her beliefs about the true demand state θ according to Figure 2, which is illustrated as follows:*

- *If the price difference between the first-period prices is less than 2Δ , where $\Delta = \frac{\alpha_H - \alpha_L}{4\gamma}$, i.e., $z_p^\theta = \frac{|p_{A,1}^\theta - p_{B,1}^\theta|}{2\Delta} \leq 1$, then, irrespective of the demand shock, both seller A and seller B always learn the true demand state, i.e., $\theta_i = \theta$ for both $i \in \{A, B\}$.*
- *If the price difference is larger than 2Δ , i.e., $z_p^\theta > 1$, then, seller i 's learning depends on the true demand state (i.e., whether $\theta = H$ or $\theta = L$) and the magnitude of the first-period demand shock, i.e.,*

$$\theta_i = \begin{cases} H & \text{if } \epsilon_1 \leq \frac{2\Delta}{p_{A,1}^\theta - p_{B,1}^\theta} \text{ and } \theta = H \\ L & \text{if } \epsilon_1 \geq 1 - \frac{2\Delta}{p_{A,1}^\theta - p_{B,1}^\theta} \text{ and } \theta = L \\ \bar{\theta} & \text{otherwise,} \end{cases}$$

The above proposition¹² clarifies that a seller's ability to learn the true demand state depends on the magnitude of the first-period price difference and demand shock. As shown in Figure 2, if the first-period prices are sufficiently close to each other (i.e., $z_p^\theta \leq 1$), both sellers are *guaranteed* to learn the true demand state through individual self-exploration; i.e., they can do so on their own. The rationale behind this is that the random noise gets amplified in the price difference. Therefore, if the price difference exceeds a certain threshold, it becomes difficult (with a smaller probability) to determine whether the observed demand is driven by the true demand state or random demand noise. If the magnitude of the first-period price difference is sufficiently large, then sellers can infer the true demand state depending on the magnitude of demand shock and true demand state. Namely, sellers can infer the true demand state only when their individual demand realization is at an extreme: either the demand shock is sufficiently low (i.e., $\epsilon_1 \leq \frac{2\Delta}{p_{A,1}^\theta - p_{B,1}^\theta}$) or the demand shock is sufficiently high (i.e., $\epsilon_1 \geq 1 - \frac{2\Delta}{p_{A,1}^\theta - p_{B,1}^\theta}$). This makes sense because when the price difference is large, the demand noise also becomes significant. Therefore, each seller needs to observe a signal that is sufficiently strong to differentiate between the high demand state and the low demand state.

The above observation has an important implication for the pricing policy of the sellers during the first period. On the one hand, if sellers A and B want to learn (i.e., *explore*) the true demand state by observing

¹² Proof in Appendix Section 3

only their own sales, they have to converge their prices closer to each other, resulting in a smaller $z_p^\emptyset$. On the other hand, the desire to *exploit* their first-period profit triggers price competition, compelling sellers to differentiate their products by widening the price gap. This creates conflicting incentives for the first-period prices and introduces an exploration and exploitation trade-off for the sellers. The following proposition fully characterizes the equilibrium first-period prices under this exploration-exploitation trade-off:

PROPOSITION 4. *The platform can induce both sellers not to subscribe by setting subscription fee $\phi^\emptyset = \max\{S_B^\emptyset, S_A^\emptyset\}$, where $S_B^\emptyset = (1 - \beta)[\pi_{CB}^B - \pi_{CB}^\emptyset]$ and $S_A^\emptyset = (1 - \beta)[\pi_{CA}^A - \pi_{CA}^\emptyset]$.¹³ Sellers A and B opt not to subscribe, and offer the following prices for the first period:*

$$p_{A,1}^\emptyset = \bar{p}^\emptyset + z_p^\emptyset \Delta, \quad \text{and} \quad p_{B,1}^\emptyset = \bar{p}^\emptyset - z_p^\emptyset \Delta$$

where $\Delta = \frac{\alpha_H - \alpha_L}{4\gamma}$, and $(\bar{p}^\emptyset, z_p^\emptyset)$ is characterized in the Appendix [Section 4](#). At the end of the first period, the sellers observe the sales and update their own beliefs according to Proposition 3. Two scenarios may occur: (i) If the sellers learn the demand, i.e., $\theta = H$ or $\theta = L$, then they set the following equilibrium second-period prices:

$$p_{A,2}^\emptyset = \frac{\gamma + (2 + \gamma)\alpha_A(\theta)}{(2 + \gamma)(2 + 3\gamma)}, \quad \text{and} \quad p_{B,2}^\emptyset = \frac{\gamma + (2 + \gamma)\alpha_B(\theta)}{(2 + \gamma)(2 + 3\gamma)}$$

(ii) If the sellers do not learn the demand, i.e., $\theta = \bar{\theta}$, they set the following equilibrium second-period prices:

$$p_{A,2}^\emptyset = \frac{1}{2(2 + \gamma)} + \frac{\alpha_H + \alpha_L - 1}{2(2 + 3\gamma)}, \quad \text{and} \quad p_{B,2}^\emptyset = \frac{1}{2(2 + \gamma)} - \frac{\alpha_H + \alpha_L - 1}{2(2 + 3\gamma)}.$$

4.3. Characterization of Equilibrium

Now, we are ready to analyze the subscription decision of the platform. By comparing Propositions 2 and 4, we can characterize the full equilibrium. Leaving the details to the Appendix, we characterize the equilibrium subscription decision of the platform and equilibrium prices and belief-updating of the sellers in the following Proposition 5:

PROPOSITION 5. *The equilibrium is divided into four regions, as summarized below and illustrated in Table ??:*

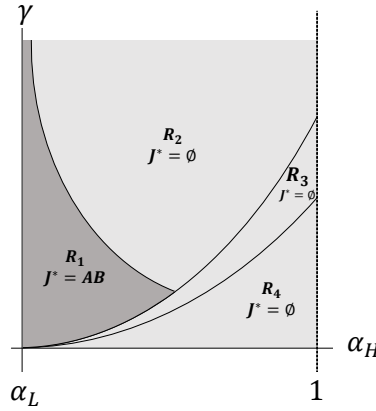
- *Region R_1 : Sellers are induced to subscribe to the information-sharing service and set their first-period prices accordingly. The platform observes sales, learns the true demand state, and shares this information with the sellers.*
- *Region R_2 : The platform sets the subscription fee high enough to discourage sellers from subscribing. Sellers set their first-period prices and update their beliefs based solely on their own demand observations.*

¹³ Any subscription fee $\phi \in [\max\{S_B^\emptyset, S_A^\emptyset\}, +\infty)$ can induce both sellers not to subscribe to CIS.

- *Regions R_3 and R_4 : Sellers do not require a subscription to learn the true demand state. They set their first-period prices and always infer the true demand by observing their own sales.*

The corresponding equilibrium subscription fees, prices, and belief-updating rules are characterized in Appendix and depicted in Figure 6 and the Proof is available in Appendix Section 5.

Figure 3 Characterization of Equilibrium Information Regimes



As characterized in Proposition 5, the full equilibrium is categorized into four regions depending on the value of system parameters. To gain a deeper insight into the equilibrium analysis, we can evaluate each region by defining a ratio denoted by $n = \frac{\alpha_H - \alpha_L}{\alpha_H + \alpha_L - 1}$. Delegating the details to the Appendix, this ratio can be shown to equal the spread between the demand states under the a priori distribution¹⁴ divided by the difference between the sellers' average demands. Hence, it can be referred to as the *signal-to-noise ratio* (SNR)¹⁵. Since the SNR (n) increases in α_H , we can rank the regions by their SNR values: region R_1 corresponds to the lowest SNR, region R_2 to a medium SNR, and regions R_3 and R_4 to the highest SNR. Using this measure, we can assess each region as follows:

- **Region R_1 :** The SNR is extremely small, indicating only a narrow gap between the two demand states. In this setting, sellers find it difficult to identify the true demand state on their own. It is therefore up to the platform to determine whether it is beneficial to induce them to subscribe to the information-sharing service (CIS), a decision that involves a trade-off between first- and second-period outcomes. Inducing both sellers to subscribe to CIS benefits the platform in the second period: by sharing information, the platform enables sellers to fully learn the demand state at the end of the first period, allowing them to set more efficient prices and ultimately increase their second-period profits. The impact on first-period profits, however, is less straightforward. If both sellers know they will receive updated beliefs before the second-period price competition, they have no incentive to bring their prices closer together in the first period. This reduces price

¹⁴ A technical proof of this equality is provided in the Appendix

¹⁵ A similar term is employed by Hansen et al. (2021) to demonstrate how long-run prices depend on informational value.

competition and lowers the platform's first-period payoff. Nevertheless, the gain in the second-period payoff from more efficient pricing outweighs this loss. Accordingly, in region R_1 , the platform has an incentive to set the subscription fee low enough to encourage both sellers to subscribe to the information-sharing service.

- Regions R_2 , R_3 , and R_4 : The SNR is sufficiently large for sellers to potentially learn the true demand state without subscribing. They can do so via one of two strategies: (i) bring their prices closer together (i.e., set $z_p \leq 1$), which - per Proposition 3 - allows both sellers to always learn the true demand state, or (ii) maintain distinct prices in the first period (i.e., set $z_p > 1$) and infer the true demand state only probabilistically. The first strategy intensifies competition, which benefits the platform during the first period, while the second may result in suboptimal second-period prices due to the possibility that the demand state remains unidentified, which hurts the platform during the second period. Nevertheless, we can show that regardless of which strategy sellers choose, it is always optimal for the platform not to share demand information. Hence, in Regions R_2 , R_3 , and R_4 , the platform has no incentive to induce sellers to subscribe to CIS. From the sellers' perspective, however, there is a trade-off between these two strategies, which depends on the level of γ , as explained below:

- Region R_2 : In this region, γ is relatively high, which increases the degree of competition between sellers. Therefore, the sellers prefer to keep their first-period prices different ($z_p > 1$). Consequently, in some cases, the sellers cannot determine the true demand state by the end of the first period and must rely on their a priori beliefs when setting second-period prices.

- Region R_3 : With sufficiently low γ , the cost of bringing prices closer is smaller due to weaker competitive pressure. Sellers, therefore, set their prices close enough ($z_p = 1$) to fully eliminate noise and always learn the true demand state.

- Region R_4 : The SNR is already very high, so sellers set $z_p < 1$, which enables them to identify the true demand state from their own demand observations without having to adjust their first-period prices.

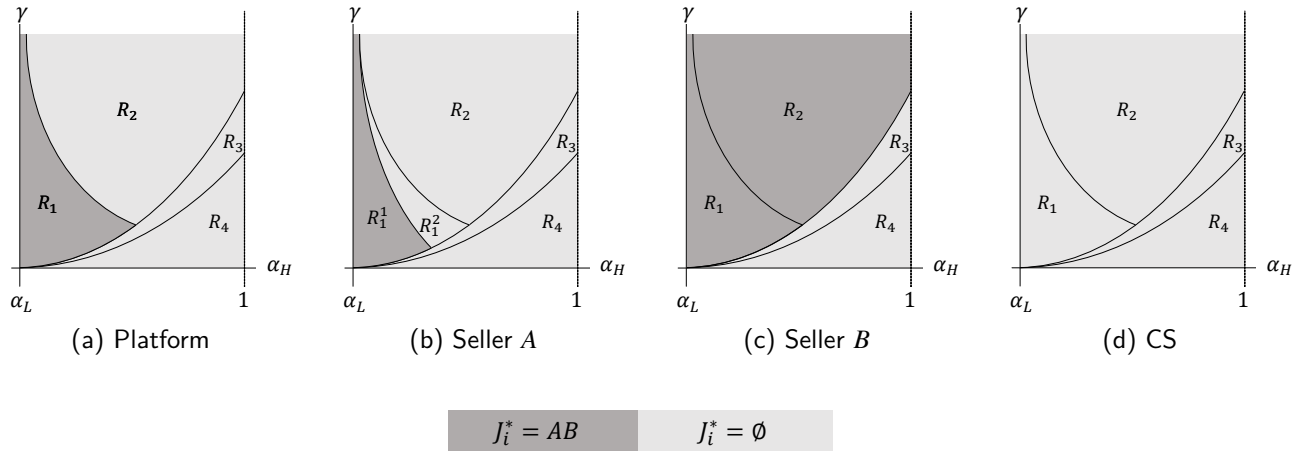
4.4. The Impact of Equilibrium on Sellers' Payoffs and Consumer Surplus

Up to this point, our analysis has focused on the platform's perspective. Since the platform's payoff depends on the combined profits of both sellers, it can tolerate a reduction in one seller's profit as long as the overall change in total profit is positive. We now extend the analysis to examine how the equilibrium decisions affect the individual payoffs of Seller A and Seller B, as formalized in the following proposition.

PROPOSITION 6. *As shown in Figure 4, the comparison between each seller's payoff under $J = AB$ and $J = \emptyset$ scenarios results in the following characterizations:*

1. *Seller A's preference is similar to that of the platform, albeit with slightly different regions, i.e., seller A prefers $J = AB$ over $J = \emptyset$ in region R_1^1 , otherwise seller A prefers $J = \emptyset$ over $J = AB$.*
2. *Seller B prefers $J = AB$ over $J = \emptyset$ in regions R_1 and R_2 , otherwise seller B prefers $J = \emptyset$ over $J = AB$.*
3. *Consumer surplus (CS) is maximized by regime $J = \emptyset$ in all regions.*

Figure 4 Payoff comparison between regimes $J = AB$ and $J = \emptyset$



Proposition 6 characterizes the preferences of sellers and consumer surplus between regime AB and regime $\emptyset$. Do they prefer exploration mediated by the platform (i.e., regime AB) or independent self-exploration (i.e., regime $\emptyset$)? Firstly, as shown in Figure 4, seller B prefers CIS regime in both regions R_1 and R_2 . But seller A prefers CIS regime only in a subregion inside R_1 , where the SNR is very low. Note that the smaller the SNR, the more difficult it becomes for seller A to independently explore the demand state, therefore, Proposition 6 can be interpreted as follows: If seller A does have to significantly reduce the price in order to self-learn, seller A can benefit from CIS, while seller B consistently benefits from CIS in both regions R_1 and R_2 . The rationale behind this interesting result is, again, tied to the impact of the method of demand exploration on competitive prices. When sellers need to explore demand themselves, seller A ends up increasing the price, while seller B decreases it. This leads to different consequences for sellers' first- and second-period profits. For both seller A and seller B , self-exploration during the first period might not always reveal the true demand state in the second period. This is because they rely solely on their own observations and may not determine the true demand state if the actual sales differ only slightly from their expectations (for detailed information, refer to Proposition 4). However, if the sellers subscribe to CIS, they receive the true demand state from the platform, regardless of the sales realization. Consequently, both sellers are worse off with self-exploration from the perspective of second-period profits.

However, self-exploration, while it may harm both sellers in the second period, can actually benefit seller A in the first period. To understand this difference, let us first consider the CIS scenario. When the platform provides information about the true demand state, sellers have complete knowledge in the second period, leading to intense competition between seller A and seller B in the first period. Seller B tends to undercut seller A . This benefits seller B , as seller B always benefits from receiving CIS, both in the short term (i.e., first period) and in the long term (i.e., second period). However, seller A prefers the no-CIS scenario over CIS in the first period, primarily because it leads to reduced price competition in the form of less aggressive

undercutting by seller B . There is, however, a caveat. Under the no-CIS regime, seller A needs to lower her price to engage in self-exploration, which results in a reduced profit margin in the first period. If the SNR is very low (i.e., subregion R_1^I in R_I), seller A needs to significantly decrease the profit margin; thus, overall, the seller benefits from CIS subscription in the first period. In summary, in subregion R_1^I , seller A 's short-term loss from no-CIS is compensated by her long-term gain from receiving CIS.

The above analysis also has direct implications for the ongoing regulatory debate on data access in digital platforms. Recent regulations across major economic zones increasingly emphasize improving sellers' access to data generated through their own platform activities. Notably, the European Union's Digital Markets Act (DMA)¹⁶ adopts a seller-centric perspective by requiring designated platform "gatekeepers" to provide business users with continuous, high-quality access to data related to their sales and performance, without discriminatory treatment across sellers of different sizes Cabral et al. (2021). Although our model treats the information-sharing regime as a choice variable of the platform provider, the fact that smaller sellers benefit disproportionately from information sharing relative to larger sellers yields insights that are consistent with this regulatory approach.

Figure 4(d) also shows that consumer surplus is maximized under regime $\emptyset$. This outcome stems from how information-sharing regimes affect equilibrium prices, as described in Proposition 5. Between regimes AB and $\emptyset$, the latter yields equilibrium prices that are more favorable for consumers. Intuitively, when both sellers must rely solely on their own sales to update beliefs, they bring their prices closer together to improve demand learning. This leads seller B to increase her price while seller A lowers hers. Although the increase in seller B 's price reduces consumer surplus, the decrease in seller A 's price has a larger positive effect, since seller A serves a larger share of the market. The net result is an overall improvement in consumer welfare under regime $\emptyset$.

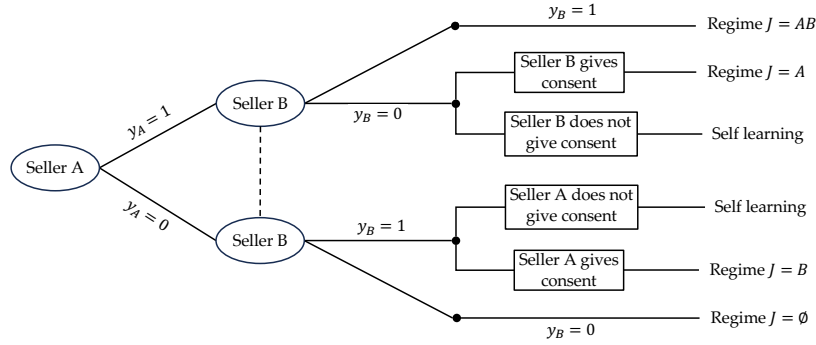
5. The Impact of Inducing Only One Seller to Subscribe to CIS

In §4, we analyzed the scenario in which the platform induces both sellers to subscribe to CIS. In this section, we extend the analysis to cases where the platform may offer the CIS subscription unilaterally to a single seller - either seller A or seller B - provided that the chosen seller agrees to pay the subscription fee. Since the platform derives demand information from the realized sales of both sellers, sharing this information with only one seller requires obtaining consent from the other. Consequently, as shown in Figure 5, unilateral information sharing is implemented only if the non-subscribing seller grants consent.

Now, we analyze the equilibrium under unilateral subscription regimes. We have two cases: Regime $J = A$ and Regime $J = B$. If the platform wants to induce CIS subscription for supplier A only, i.e., forming the

¹⁶ EU's Digital Markets Act (DMA) entered into force on November 1, 2022 and become applicable on May 2, 2023.

Figure 5 Sellers' subscription and consent decisions



subscription regime $J = A$, then the subscription fee offered by the platform should satisfy the following incentive compatibility constraints of the sellers:

$$\pi_B^A(P_B^A, P_A^A) \equiv (1 - \beta)\pi_{CIB}^A(P_B^A, P_A^A) \geq (1 - \beta)\pi_{CIB}^{AB}(P_B^{AB}, P_A^{AB}) - \phi^A \equiv \pi_B^{AB}(P_B^{AB}, P_A^{AB}) \quad (11)$$

$$\pi_A^A(P_A^A, P_B^A) \equiv (1 - \beta)\pi_{CIA}^A(P_A^A, P_B^A) - \phi^A \geq (1 - \beta)\pi_{CIA}^\emptyset(P_A^\emptyset, P_B^\emptyset) \equiv \pi_A^\emptyset(P_A^\emptyset, P_B^\emptyset) \quad (12)$$

where constraint (11) ensures that when seller A subscribes to CIS, seller B has no incentive to subscribe. Similarly, constraint (12) ensures that if seller B decides not to subscribe, seller A has no incentive to deviate and not subscribe. These two constraints together define subscription regime $J = A$ where $y_A = 1$ and $y_B = 0$. Similarly, the following constraints should be met if the platform intends to induce only seller B to subscribe to CIS, hence forming regime $J = B$:

$$\pi_B^B(P_B^B, P_A^B) \equiv (1 - \beta)\pi_{CIB}^B(P_B^B, P_A^B) - \phi^B \geq (1 - \beta)\pi_{CIB}^\emptyset(P_B^\emptyset, P_A^\emptyset) \equiv \pi_B^\emptyset(P_B^\emptyset, P_A^\emptyset) \quad (13)$$

$$\pi_A^B(P_A^B, P_B^B) \equiv (1 - \beta)\pi_{CIA}^B(P_A^B, P_B^B) \geq (1 - \beta)\pi_{CIA}^{AB}(P_A^{AB}, P_B^{AB}) - \phi^B \equiv \pi_A^{AB}(P_A^{AB}, P_B^{AB}) \quad (14)$$

Before characterizing the equilibrium, we first show that inducing subscription to seller B (i.e., the seller with the smaller market size) is not feasible. This issue stems from conflicts between the subscription incentives. On one hand, constraint (13) represents the maximum allowable value of the subscription fee to ensure that seller B retains incentives to subscribe. On the other hand, constraint (14) signifies the minimum value of a subscription fee needed to prevent seller A from subscribing. These two conditions together establish bounds on the subscription fee when inducing $J = B$, namely: $\pi_{CIA}^{AB} - \pi_{CIA}^B < \phi^B < \pi_{CIB}^B - \pi_{CIB}^\emptyset$. However, the lower bound is greater than the upper bound. Therefore, as characterized below, implementing the subscription regime $J = B$ is not possible.

PROPOSITION 7. *Regime $J = B$ is not implementable, i.e., the platform cannot induce only seller B to subscribe.*

Next, we consider the case where the platform aims to induce only seller A to subscribe to CIS. At the end of the first period, seller A receives an updated belief from the platform, while seller B relies solely on her own sales. Our analysis reveals that seller B 's belief updating follows the same process as the one

characterized under the no-CIS scenario (regime $J = \emptyset$). In other words, as outlined in Proposition 4, seller B updates her beliefs as follows:

$$\theta_B = \begin{cases} H & \text{if } \epsilon_1 \leq \frac{2\Delta}{p_{A,1} - p_{B,1}} \text{ and } \theta = H \\ L & \text{if } \epsilon_1 \geq 1 - \frac{2\Delta}{p_{A,1} - p_{B,1}} \text{ and } \theta = L \\ \bar{\theta} & \text{otherwise,} \end{cases} \quad (15)$$

where $\Delta = \frac{\alpha_H - \alpha_L}{4\gamma}$. Since ϵ_1 is bounded between 0 and 1, the updating rule implies that seller B can learn the true demand state when the first-period price difference between seller A and B is sufficiently small. In §4, we showed that both sellers have an incentive to bring their prices closer together to reduce demand noise and facilitate demand learning. However, under regime $J = A$, seller A is induced to subscribe to CIS and thus receives the true demand state directly from the platform, while seller B must still rely on her first-period sales to update her beliefs on the true demand state. In this case, seller B continues to have an incentive to align her price with seller A 's, but seller A no longer benefits from doing so. In fact, as we characterize below, seller A now has an incentive to increase the first-period price difference, which in turn makes demand learning more difficult for seller B . We refer to this as signal-jamming and analyze its impact on equilibrium prices.

PROPOSITION 8. *The equilibrium under regime A is characterized as follows:*

- *The platform can induce only seller A to subscribe by offering $\phi^A = (1 - \beta) [\pi_{CIA}^A(P_A^A, P_B^A) - \pi_{CIA}^\emptyset(P_A^\emptyset, P_B^\emptyset)]$.*
- *Sellers A and B offer the following prices in the first period:*

$$p_{A,1}^A = \bar{p}_1^A + z_p^A \Delta, \quad \text{and} \quad p_{B,1}^A = \bar{p}_1^A - z_p^A \Delta$$

where $\bar{p}_1^A = \frac{1}{2(2+\gamma)} + \frac{\gamma(\alpha_H - \alpha_L)}{16(1+\gamma)(2+\gamma)(z_p^A)^2}$, z_p^A is described in the Appendix and $\Delta = \frac{\alpha_H - \alpha_L}{4\gamma}$.

– *After the first period sales, seller A receives the true demand state from the platform and seller B updates her own beliefs according to Equation 15.*

- *In the second period, the following equilibrium prices are set if seller B learns the true demand state:*

$$p_{A,2}^A = \frac{\gamma + (2 + \gamma)\alpha_A(\theta)}{(2 + \gamma)(2 + 3\gamma)}, \quad \text{and} \quad p_{B,2}^A = \frac{\gamma + (2 + \gamma)\alpha_B(\theta)}{(2 + \gamma)(2 + 3\gamma)}$$

and the following equilibrium prices are set if seller B does not learn:

$$p_{A,2}^A = \frac{\gamma + (2 + \gamma)\alpha_A(\theta)}{(2 + \gamma)(2 + 3\gamma)} + \frac{\gamma[\alpha_H + \alpha_L - 2\alpha_A(\theta)]}{4(1 + \gamma)(2 + 3\gamma)}, \quad \text{and} \quad p_{B,2}^A = \frac{4(1 + \gamma) - (2 + \gamma)(\alpha_H + \alpha_L)}{2(2 + \gamma)(2 + 3\gamma)}.$$

The first part of Proposition 8 establishes that inducing subscription from seller A , i.e., the seller with the larger market size, is feasible provided the subscription fee ϕ^A is chosen appropriately. Specifically, the fee must be high enough to deter seller B from subscribing to CIS, yet low enough to ensure that seller A finds subscription profitable. By setting ϕ^A accordingly, the platform can satisfy the incentive compatibility constraints of seller A and B . The second part of Proposition 8 characterizes the first-period equilibrium prices under regime $J = A$. Since seller A receives updated demand beliefs from the platform, she has no incentive to engage in exploration and instead focuses solely on exploitation in the first period. This behavior has two key implications. First, it increases the price gap between sellers A and B . As characterized in the

below Proposition 9, we compare the first-period equilibrium prices across all three regimes and demonstrate that the price difference between A and B is larger under regime A than under regime $\emptyset$. This wider gap between the first-period prices generates a signal-jamming effect: the greater the initial price difference, the less likely seller B can update her beliefs about the true demand state. Second, as seller A raises her price to hinder seller B 's learning, seller B responds by increasing her own price. Consequently, the average price level under regime A exceeds that characterized under both the full-information regime ($J = AB$) and the no-information regime ($J = \emptyset$).

PROPOSITION 9. *The first-period equilibrium prices under the three regimes satisfy:*

- *The ranking of first-period price differences between the sellers is $z_p^{AB} > z_p^A > z_p^\emptyset$.*
- *Let $\bar{p}^J = \frac{p_A^J + p_B^J}{2}$ denote the average first-period price in regime J . Then $\bar{p}^A > \bar{p}^{AB} = \bar{p}^\emptyset$.*

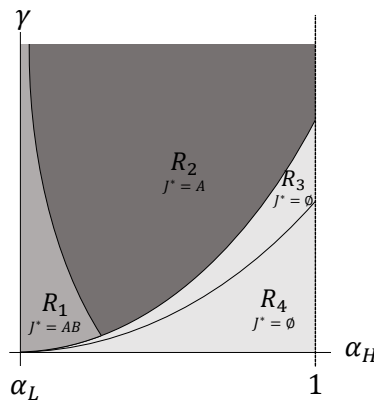
Note that the above proposition not only reveals the signal-jamming behavior but also has implications for the platform's preference in implementing regime A , regime AB or regime $\emptyset$. By comparing the platform's payoffs under these three regimes, we can now characterize the equilibrium as follows:

PROPOSITION 10. *The equilibrium is divided into four regions, as summarized below:*

- *Region R_1 : Both seller A and seller B are induced to subscribe to the information-sharing service.*
- *Region R_2 : Only seller A is induced to subscribe to the information-sharing service.*
- *Regions R_3 and R_4 : Neither seller A nor seller B is induced to subscribe to the information-sharing service.*

The corresponding equilibrium subscription fees, first- and second-period prices, and belief-updating rules are characterized in Appendix for each region depicted in Figure 6.

Figure 6 Characterization of Equilibrium Information Sharing Regimes



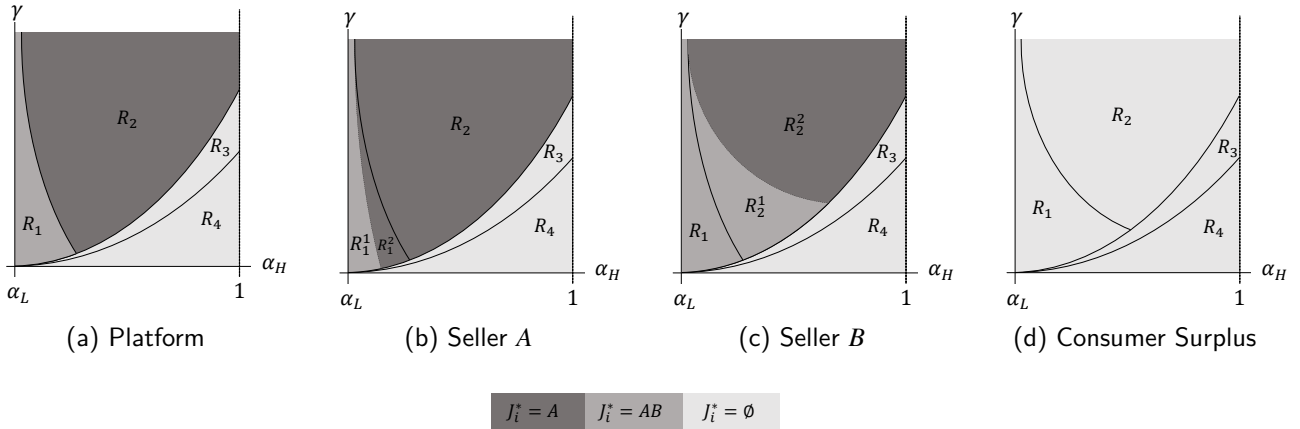
The comparison between the equilibria characterized in Proposition 10 and Proposition 5 reveals both similarities and differences. First, similar to Proposition 5, the equilibria in Proposition 10 also consist of

four distinct regions, although regions R_1 and R_2 differ slightly from those in Proposition 5. Second, in all regions except R_2 , the same regime is implemented in both propositions. The key distinction arises in region R_2 , where regime $\emptyset$ in Proposition 5 is replaced by regime A in Proposition 10.

This difference stems from the signal-jamming behavior of seller A toward seller B . In region R_2 under Proposition 5, the platform benefits from not sharing demand information, as the sellers are incentivized to align their prices more closely, thereby intensifying first-period price competition. In contrast, under regime A , the platform can selectively set the subscription fee to incentivize only seller A to subscribe to the information-sharing service. With access to information, seller A increases her price to jam seller B 's demand learning, prompting seller B to increase her own price in response. As a result, both sellers charge higher prices. Even though this results in lower demand for both sellers, the increase in equilibrium prices ultimately increases the platform's total revenue.

Although closed-form expressions for equilibrium prices in both periods can be derived, the complicated nature of these closed-form expressions makes direct comparisons of seller profits and consumer surplus across regions intractable. To address this, we provide a graphical comparison of profits and consumer surplus in Figure 7.

Figure 7 Payoff comparison in Full Equilibrium, $\alpha_L = 0.65$



Since the platform's preferences have already been discussed, we now restrict attention to the effects of different regions on the profits of sellers A and B , as well as consumer surplus:

- **seller A 's preference:** As shown in Figure 7(b), seller A 's preferences are generally aligned with those of the platform. This is intuitive: the platform shares demand information exclusively with seller A , which benefits her in two ways. During the first period, seller A can sustain higher prices, and at the start of the second period, she always has access to the true demand state, whereas seller B may not. This ensures that seller A maintains a competitive advantage over seller B in the second period. Indeed, our numerical analysis

confirms that seller A prefers regime A over regime AB in region R_1 , since under regime AB the true demand information is shared with both sellers, while under regime A it is available only to seller A . Thus, seller A preserves a competitive edge in regime A , making it more favorable than regime AB in region R_1 .

- **seller B 's preference:** Figure 7(c) reveals a more surprising result: within a subregion of R_2 , seller B also prefers regime A over regime $\emptyset$. Although regime A puts seller B at a competitive disadvantage in the second period - since seller A holds exclusive access to the demand state - seller B nonetheless gains in the first period. The reason is that seller A 's signal-jamming behavior increases not only her own equilibrium price but also that of seller B . The resulting increase in seller B 's first-period revenue can outweigh the loss from reduced learning in the second period. In summary, seller B prefers regime AB over regime A only when both the signal-to-noise ratio (SNR) and the degree of competition are very low - corresponding to region R_1 and the left portion of region R_2 (i.e., lower α_H and lower γ). In contrast, she prefers regime A when competition is more intense and the SNR is relatively low - corresponding to the right portion of region R_2 . This right-hand side of region R_2 is precisely where the signal-jamming effect is strongest, leaving seller B worse off in relative terms in the second period, yet still benefiting her due to higher equilibrium prices in the first period.

- **Consumer surplus:** Figure 7(d) shows that the consumer surplus is maximized under regime $\emptyset$ even after we add regime A into the consideration set. The rationale behind this is due to the effect of information-sharing regimes on equilibrium prices, as characterized in Proposition 9. Among all regimes, regime $\emptyset$ yields the most favorable prices from the consumers' perspective. Intuitively, when both sellers must rely solely on their own sales to update beliefs, they bring their prices closer together to improve demand learning. This basically implies that seller B increases her price, while seller A decreases her price. Although the increase in seller B 's price reduces consumer surplus, the decrease in seller A 's price generates a larger positive effect, since seller A serves a larger market. As a result, the net impact is an overall improvement in consumer surplus under regime $\emptyset$.

We conclude this section by noting that the information-sharing regimes characterized here are also consistent with practices observed in digital markets. In particular, in the United States and Canada, platforms commonly allow sellers to opt into analytics services that process seller-level data and generate insights that may be informative to other sellers, including competitors. For example, platforms such as Amazon offer opt-in analytics tools (e.g., Brand Analytics(Amazon 2025)) that use seller-level sales data to generate pricing, demand, or inventory recommendations. Platforms may also provide insights derived from aggregated activity across multiple sellers, such as market-basket analysis identifying products that are frequently purchased together. As shown in this section, such information-sharing arrangements can arise endogenously in equilibrium.

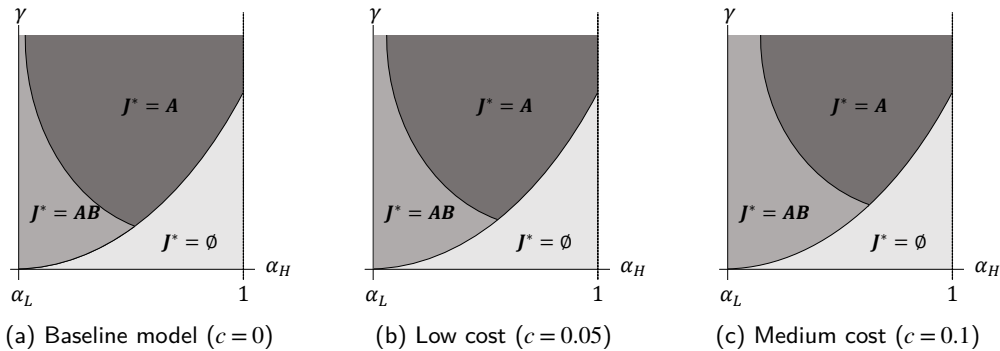
6. Robustness Check of Analytical Results

While §5 analytically established the impact of competitive demand exploration-exploitation on pricing strategies and payoffs, the results were obtained under the following assumptions required for tractability: (i) $c_A = c_B = 0$, i.e., the marginal cost is homogeneous and zero for both sellers, (ii) $\alpha_A(\theta) + \alpha_B(\theta) = 1$ for $\theta \in \{L, H\}$, i.e., the total market size under both demand states is normalized to be equal to 1, (iii) the noise in demand for both periods is distributed uniformly between 0 and 1, i.e., $\epsilon_t \sim U[0, 1]$, and (iv) ϵ_1 and ϵ_2 are independent. It turns out that relaxing any of these assumptions makes it challenging to characterize equilibrium decisions and, more importantly, to analytically establish the effects of different CIS arrangements. In this section, we numerically investigate whether or not the managerial insights presented in §4.1 and §5 remain valid if we relax the above assumptions. While our numerical study can validate the robustness of all our results, below we focus on the insights about how heterogeneous marginal costs and state-dependent total demand affect the platform's incentive to provide CIS.

6.1. Heterogeneous Marginal Costs

In this section, we use the same model framework as in §3 and consider the case where the marginal costs for sellers are heterogeneous. It is reasonable to assume that $c_A > c_B$, since seller A 's market share is larger than seller B 's market share for both demand states. For expositional brevity, we set $c_A = c > 0$ and $c_B = 0$. By varying the value of c , we focus on the impact of c on the equilibrium regions. Specifically, we consider three cases: the baseline model ($c = 0$), a medium cost ($c = 0.05$), and a high cost ($c = 0.1$). We pick these values because when c exceeds 0.1, seller A finds it unprofitable to remain in the market, which leads to a monopolistic outcome for seller B and thus trivial results for our model. The results are presented in Figure 8:

Figure 8 Comparison of the equilibrium information regimes under various marginal cost scenarios



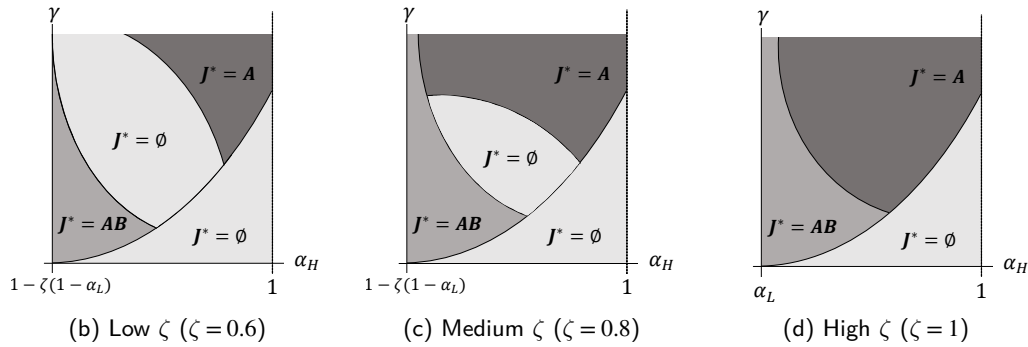
Recall that in §4 and §5 we show that under both regime $J = \emptyset$ and regime $J = A$, sellers narrow the first-period price gap in order to increase the probability of learning. This implies that seller A must decrease her first-period price, while seller B must increase her price. However, when seller A faces a positive marginal cost, her ability to reduce the price in pursuit of learning is limited. Consequently, both regimes $J = \emptyset$ and

$J = A$ become more costly for the platform. As illustrated in Figure 8(a), our numerical results demonstrate that as marginal cost c for seller A increases, regime AB becomes relatively more advantageous for the platform compared to both regime $J = \emptyset$ and regime $J = A$.

6.2. State-dependent Total Demand

In the baseline model, we normalized the total demand under both demand states to be equal to 1. In this section, we relax this assumption. That is, for the high-demand state, we still normalize the total market size to 1, but for low demand state, we normalize it to be equal to ζ , where $\zeta \in (0, 1]$. Similar to §6.1, we only focus on the impact of ζ on the comparison between equilibrium information-sharing regimes. We consider three values for ζ : low ζ (i.e., $\zeta = 0.6$), medium ζ (i.e., $\zeta = 0.8$) and high ζ (i.e., $\zeta = 1$). The results are presented in Figure 9.

Figure 9 Comparison of the equilibrium information regimes under various state-dependent demand scenarios



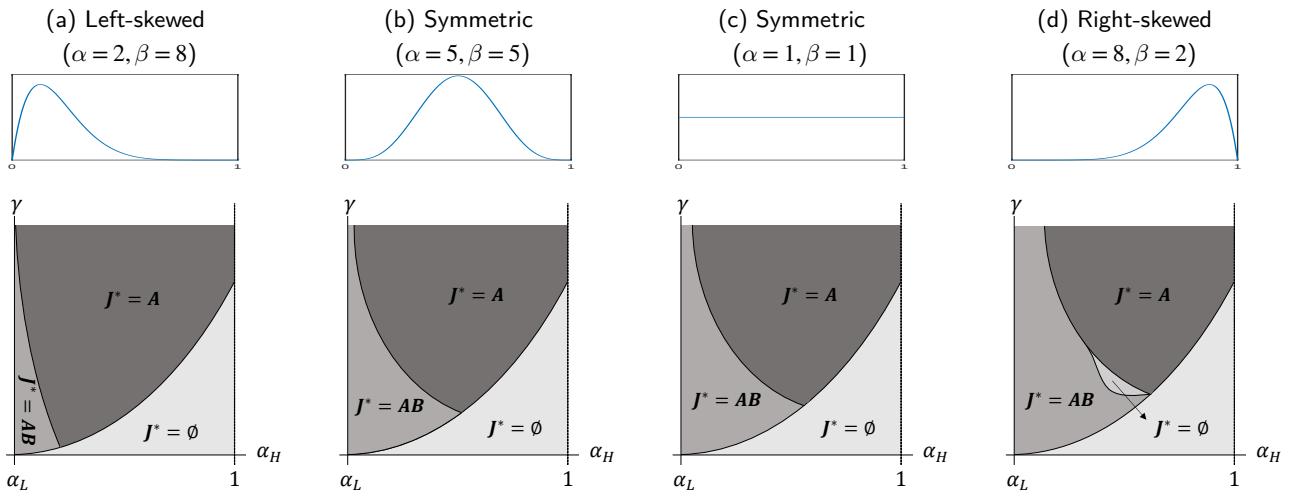
Unlike in §6.1, the state-dependent aggregate demand extension changes the possible learning cases. When each seller self-explores the demand state by observing only their own sales, it is possible that seller A learns the true demand state while seller B does not. Moreover, by setting a price sufficiently far from the competitor's, seller A can ensure that she always learns the true demand state, regardless of demand realization, whereas seller B cannot.

This new possibility has two implications. First, unlike in §4.2, seller A no longer needs the platform's demand information to engage in signal-jamming against seller B in the first period. Second, this dynamic leads to price collusion, which raises the average equilibrium price in the first period. Consequently, regime $J = \emptyset$ emerges as the equilibrium in Figure 9, since it is preferred from the platform's perspective. However, as α_H increases, sustaining signal-jamming under regime $J = \emptyset$ becomes harder: the growing price gap between sellers A and B makes the signal-jamming region increasingly small. Hence, for higher values of α_H , the platform prefers to intervene and implement regime $J = A$.

6.3. Non-uniform noise distribution

In the baseline model, we assumed that ϵ_t follows a uniform distribution on $[0, 1]$. In this section, we relax this assumption and consider non-uniform cases by modeling ϵ_t with a parametrizable beta distribution, $\epsilon_t \sim \text{Beta}(\alpha, \beta)$. Since the beta distribution is bounded between 0 and 1, it provides a flexible way to capture different distributions between 0 and 1. By varying α and β , we examine left-skewed ($\alpha = 2, \beta = 8$), right-skewed ($\alpha = 8, \beta = 2$), and two symmetric distributions: the uniform case ($\alpha = 1, \beta = 1$) and a more concentrated symmetric case ($\alpha = 5, \beta = 5$). Figure 10 presents the results.

Figure 10 Comparison of the equilibrium information regimes under various distributions



Note that the uniform distribution ($\alpha = 1, \beta = 1$) corresponds to our baseline model. For benchmarking, we replicate this case in panel (c) of 10 above and focus our discussion on how equilibrium regimes change as the distributional shape varies. Increasing both α and β from 1 to 5 preserves the mean of ϵ_t at $\frac{1}{2}$ but reduces its variance. Comparing panels (b) and (c) of 10, we observe that when ϵ_t becomes more concentrated around its mean, the equilibrium region associated with regime AB shrinks. This is intuitive: as demand noise diminishes, the need for information sharing between both sellers declines.

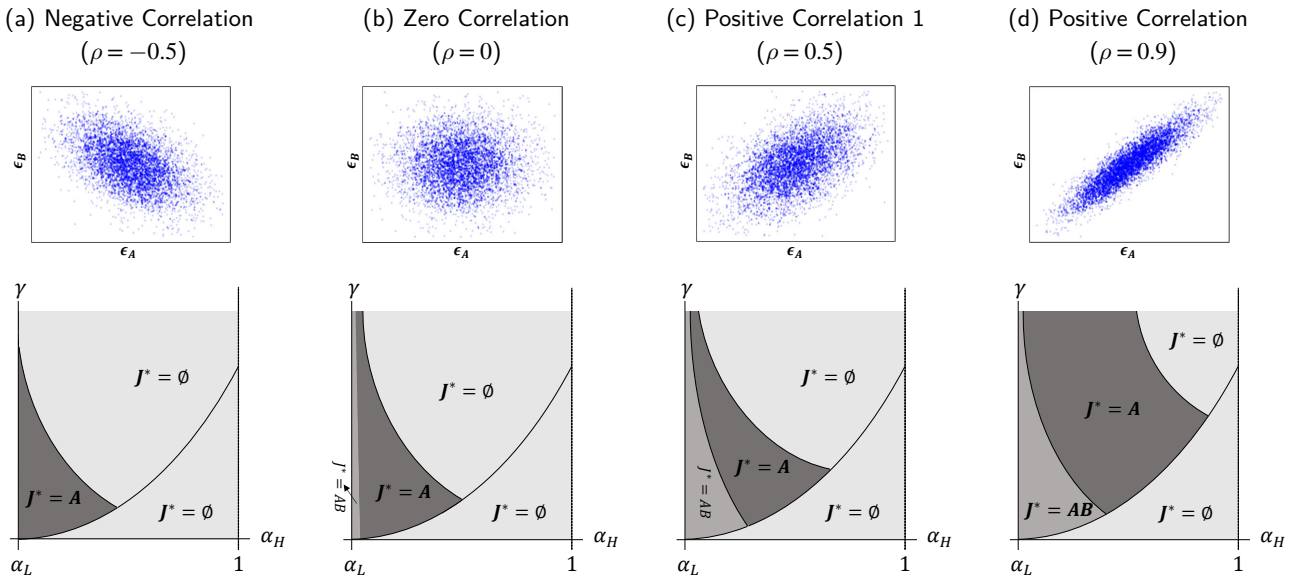
We next consider the two asymmetric cases. In the left-skewed case (panel (a) of 10), the mean of ϵ_t shifts toward 0, which reduces demand noise. Consequently, the equilibrium region for regime AB contracts further, as sharing information becomes less valuable. In contrast, in the right-skewed case (panel (d) of 10), the mean shifts toward 1, amplifying demand noise. This has two effects. First, the equilibrium region for regime AB expands. Second, unlike the baseline model where regime A always dominates regime $\emptyset$, we now observe that the platform sometimes prefers regime $\emptyset$ over both A and AB . The intuition is that although the mean shifts toward 1, the distribution is also more concentrated than under the uniform case, which reduces the overall uncertainty. Under these conditions, if the platform induces regime $\emptyset$, sellers can still learn the true

demand state from their realized sales. As a result, regime $\emptyset$ emerges as an equilibrium, particularly when α_H is moderate and γ is relatively low, both of which enhance sellers' ability to learn demand.

6.4. Correlated noise distribution

We now analyze how correlations between ϵ_A and ϵ_B (random shocks faced by seller A and seller B , respectively) shape the equilibrium regimes. To capture varying degrees of correlation, we model ϵ_A and ϵ_B using a truncated multivariate normal distribution with mean centered at 0.5 and variance-covariance matrix Σ given by $\Sigma = \sigma^2 \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$, where σ^2 captures the variance around the mean and $\rho \in [-1, 1]$ controls the correlation between ϵ_A and ϵ_B . As ρ approaches -1 (resp. $+1$), the demand noise between seller A and seller B becomes more negatively (resp. positively) correlated. We consider three values: $\rho = -0.5$, $\rho = 0$, and $\rho = +0.5$. We also present $\rho = 0.9$ as a benchmark for the baseline model. The equilibrium regions are displayed in Figure 11.

Figure 11 Comparison of the equilibrium information regimes under various correlated distributions



As shown in Figure 11, equilibrium region $J = \emptyset$ emerges as the equilibrium information-sharing regime in the upper-right part of the parameter space (where both α_H and γ are high). The rationale for this outcome lies in the changes to the demand learning stage when the degree of correlation between demand shocks decreases. Specifically, as correlation weakens, the signals observed by the sellers diverge, leading to asymmetric learning outcomes.

Interestingly, when demand correlation is low or negative, two new cases arise: (i) seller A learns the true demand state while seller B does not, and (ii) seller B learns while seller A does not. Both cases induce signal-jamming behavior between the sellers. However, the consequences for the platform differ: when seller

A engages in signal-jamming, it raises prices, which benefits the platform, whereas when seller B engages in signal-jamming, it depresses prices, which harms the platform.

In the upper-right region of the parameter space (where α_H is substantially larger than α_L), the platform benefits on average from seller A 's signal-jamming. Consequently, regime $J = \emptyset$ becomes the platform's preferred equilibrium. By contrast, as α_H decreases, seller B 's signal-jamming becomes too costly, prompting the platform to intervene and induce information sharing with seller A .

7. Conclusion

Inarguably, the exponential growth in online marketplaces will attract not only more consumers but also more *sellers*, creating a win-win scenario for all stakeholders. In this paper, we focus on the Competitive Intelligence Services (CIS) that a platform can offer to its sellers. These services leverage the platform's ability to learn about the unknown price-demand relationship faster than the sellers themselves. We examine the demand exploration-exploitation trade-off faced by two sellers, competing over two periods, with varying market shares on an online platform. Each seller updates their estimate for the price-demand relationship based on individual sales observations. However, by observing the sales of both sellers, the platform excels at exploring demand, turning this demand information into valuable competitive business intelligence. As a result, the platform can mediate the demand exploration of the sellers in the first period by influencing their decision to subscribe or not to subscribe to CIS. We analytically establish the specific conditions that determine the optimal approach to demand exploration and the optimal CIS arrangement in online marketplaces. First, we characterize whether the platform should induce sellers to subscribe to CIS or not. Second, if the platform undertakes demand exploration, should it induce both sellers to subscribe to CIS, or just one of them? Finally, if the platform induces only one of the sellers to subscribe, which one should it be?

To address these research questions, we characterize closed-form expressions for the associated equilibrium prices and payoffs for the platform and the two sellers under all demand-exploration scenarios. From the platform's perspective, our analysis suggests that by offering CIS to one or both sellers, the platform alters the nature of pricing decisions in the second period. More importantly, it also changes the nature of self-exploration and, consequently, the equilibrium pricing decisions in the first period. This transformation may have either a beneficial or detrimental impact on the profitability of both the sellers and the platform.

We also establish that the signal-to-noise ratio (SNR) of the random demand faced by the sellers is a key factor in determining the optimal CIS subscription strategy in a competitive environment. Specifically, it is optimal for the platform to offer CIS to both sellers when the SNR is low. This is because a low SNR makes self-exploration difficult for each seller in the first period. Consequently, the poor knowledge of the price-demand relationship in the second period leads to lower payoffs for both the sellers and the platform. This result appears to be rather robust and remains valid under general assumptions about the degree of demand substitution between sellers and heterogeneity in their marginal costs. In fact, as the degree of

demand substitution and heterogeneity in their marginal costs increase, the provision of CIS by the platform becomes even more critical. Finally, when the SNR is high, the identity of the seller who subscribes to the CIS determines the level of price competition during the first period. Our results indicate that inducing CIS subscription solely for the small seller is not incentive-compatible. However, the larger seller benefits when she receives the demand information. Surprisingly, our analysis shows that the small seller, as well as the platform, also gain an advantage when the demand information is shared with the opponent. This is due to price collusion triggered by the signal-jamming behavior of the larger seller. Our robustness check confirms that the larger seller engages in signal-jamming even if demand exploration is pursued independently by each seller.

Last but not least, our results highlight the inherent trade-offs faced by policymakers when regulating data sharing in digital markets. Highly restrictive regimes may limit platforms' ability to leverage data-driven efficiencies, potentially slowing market expansion, whereas overly permissive approaches risk facilitating anti-competitive behavior and disadvantaging smaller sellers and consumers. Our analysis suggests that carefully designed information-sharing rules are essential for the healthy development of digital markets. Along these lines, several important design issues remain open for future research. First, while many regulatory frameworks allow information to be shared in aggregated form, an open question concerns how such aggregation should be designed. Our model focuses on market size as the relevant information dimension, but in practice platforms may share more granular information, such as clickstream data capturing product views, add-to-cart actions, and purchases. Second, another important question is whether information-sharing regimes should be implemented on an opt-in or opt-out basis. Studying these design choices would help inform emerging regulatory frameworks that aim to account for the perspectives of multiple stakeholders. We leave these issues for future research.

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